Exploratory Data Analysis On E-Commerce Sales

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
plt.style.use('ggplot')
sns.set_palette('deep')

df=pd.read_csv(r"C:\Users\Sahil Singh\Sam_Analysis\03_Project_E-
Commerce_Data(EDA)/E-Commerce_Sales_Data_2024-25.csv")
```

Understanding Dataset

```
#Displays first 10 rows.
df.head(10)  # Dataset contains order-level sales information.
```

```
   Order ID  Order Date        Customer Name       City   Category  \
0     10001  10/19/2024         Kashvi Varty  Bangalore      Books
1     10002   8/30/2025          Advik Desai      Delhi  Groceries
2     10003   11/4/2023          Rhea Kalla      Patna    Kitchen
3     10004   5/23/2025            Anika Sen    Kolkata  Groceries
4     10005   1/19/2025          Akarsh Kaul       Pune   Clothing
5     10006    4/6/2025  Vardaniya Jayaraman       Pune  Furniture
6     10007  10/13/2023        Drishya Khare     Mumbai   Clothing
7     10008   8/17/2025            Misha Dua    Lucknow      Books
8     10009    3/7/2025          Arhaan Vala     Jaipur  Groceries
9     10010   12/2/2024        Lavanya Hayer        Goa    Kitchen

          Product Name  Quantity  Unit Price
status  \
0    Non-Fiction Ipsum         2       36294
Cancelled
1            Rice Nemo         1       42165  Shipped - Delivered to
Buyer
2          Juicer Odio         4       64876
Shipped
3        Oil Doloribus         5       37320
Cancelled
4        Kids Wear Quo         1       50037
Shipped
5      Chair Assumenda         5       40287
Shipped
6   Accessories Minima         2        3636
Shipped
7        Biography Vel         1       15885  Shipped - Delivered to
Buyer
8      Spices Expedita         1       40834
```

```
Cancelled
9  Juicer Voluptatibus         2       16686
Shipped

  Payment Mode
0   Debit Card
1   Debit Card
2  Credit Card
3          UPI
4   Debit Card
5  Credit Card
6   Debit Card
7   Debit Card
8  Credit Card
9          UPI
```

```
#Check last rows
df.tail()
```

```
      Order ID  Order Date      Customer Name       City     Category
\
4995     14996   6/25/2024   Nishith Kulkarni    Kolkata        Books

4996     14997  12/22/2024      Aaina Chander     Jaipur         Toys

4997     14998   4/15/2025       Dhanush Gara  Bangalore       Beauty

4998     14999    7/8/2024  Divyansh Malhotra    Kolkata  Electronics

4999     15000    2/4/2024       Aarush Walla        Goa     Clothing


             Product Name  Quantity  Unit Price
status  \
4995    Fiction Veritatis         3       60671
Shipped
4996           Doll Nulla         5       70048  Shipped - Delivered
to Buyer
4997       Lipstick Eaque         1       42162  Shipped - Delivered
to Buyer
4998  Smartwatch Adipisci         4       13568
Shipped
4999   Kids Wear Repellat         1       76762  Shipped - Delivered
to Buyer

     Payment Mode
4995   Debit Card
4996  Credit Card
4997   Debit Card
```

```
4998  Credit Card
4999  Net Banking
```

```
    #Check Statistical Summery
    df.describe(include='all')  #Total price distribution appears
right-skewed.
```

```
            Order ID  Order Date    Customer Name      City
Category  \
count    5000.000000        5000             5000      5000      5000

unique           NaN         730             4844        20        10

top              NaN  12/31/2023  Vritika Cherian  Guwahati     Books

freq             NaN          15                3       293       528

mean    12500.500000         NaN              NaN       NaN       NaN

std      1443.520003         NaN              NaN       NaN       NaN

min     10001.000000         NaN              NaN       NaN       NaN

25%     11250.750000         NaN              NaN       NaN       NaN

50%     12500.500000         NaN              NaN       NaN       NaN

75%     13750.250000         NaN              NaN       NaN       NaN

max     15000.000000         NaN              NaN       NaN       NaN


          Product Name     Quantity    Unit Price   status Payment
Mode
count             5000  5000.000000   5000.000000     5000
5000
unique            3835          NaN           NaN        6
5
top     Chair Adipisci          NaN           NaN  Shipped   Net
Banking
freq                 5          NaN           NaN     3031
1010
mean               NaN     2.992600  39760.904600      NaN
NaN
std                NaN     1.413133  22831.783946      NaN
NaN
min                NaN     1.000000    222.000000      NaN
NaN
25%                NaN     2.000000  20312.250000      NaN
NaN
50%                NaN     3.000000  39459.500000      NaN
```

```
NaN
75%                NaN     4.000000  59721.750000      NaN
NaN
max                NaN     5.000000  79998.000000      NaN
NaN
```

```python
#Check data_types,Null_values,Memory_usage
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 5000 entries, 0 to 4999
Data columns (total 10 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   Order ID        5000 non-null   int64
 1   Order Date      5000 non-null   str
 2   Customer Name   5000 non-null   str
 3   City            5000 non-null   str
 4   Category        5000 non-null   str
 5   Product Name    5000 non-null   str
 6   Quantity        5000 non-null   int64
 7   Unit Price      5000 non-null   int64
 8   status          5000 non-null   str
 9   Payment Mode    5000 non-null   str
dtypes: int64(3), str(7)
memory usage: 740.4 KB
```

```python
#Checks column names.
df.columns
```

```
Index(['Order ID', 'Order Date', 'Customer Name', 'City', 'Category',
       'Product Name', 'Quantity', 'Unit Price', 'status', 'Payment
Mode'],
      dtype='str')
```

Data Cleaning..

```python
#Missing Values
df.isnull().sum()  #None In this DataSet
```

```
Order ID         0
Order Date       0
Customer Name    0
City             0
Category         0
Product Name     0
Quantity         0
Unit Price       0
status           0
```

```
Payment Mode      0
dtype: int64
```

```python
# Duplicate Values
df.duplicated().sum()   # None in this dataset
```

```
np.int64(0)
```

```python
# Dropping columns
df.drop('Order ID',axis=1,inplace=True)

# Standardize the data
df.columns=df.columns.str.lower().str.strip().str.replace(' ','_')
df=df.rename(columns={'unit_price':'price'})
df.columns
```

```
Index(['order_date', 'customer_name', 'city', 'category',
'product_name',
       'quantity', 'price', 'status', 'payment_mode'],
      dtype='str')
```

```python
# Fixing Data types
# Here, Order date columns show Str
df['order_date']=pd.to_datetime(df['order_date'], format="mixed")
df['order_date'].dtype
```

```
dtype('<M8[us]')
```

```python
# Feature Engineering
df['order_month']=df['order_date'].dt.month_name()
df['order_year']=df['order_date'].dt.year
df['order_month'].head()
df['order_year'].head()
```

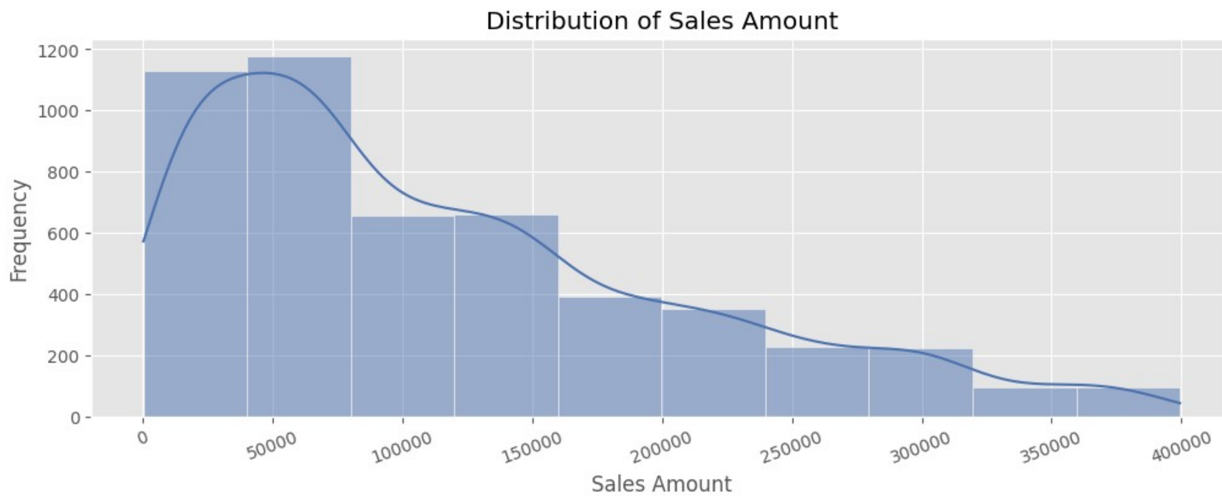
```
0    2024
1    2025
2    2023
3    2025
4    2025
Name: order_year, dtype: int32
```

```python
df['total_price']=df['price']*df['quantity']  #Important for KPIs
df['total_price'].head()
```

```
0     72588
1     42165
2    259504
3    186600
4     50037
Name: total_price, dtype: int64
```

Analysis

```
#Uninvariate Analysis

plt.figure(figsize=(12,4))
sns.histplot(df['total_price'],bins=10,kde=True)
plt.title('Distribution of Sales Amount')
plt.xlabel('Sales Amount')
plt.ylabel('Frequency')
plt.xticks(rotation=20)
plt.show()
```

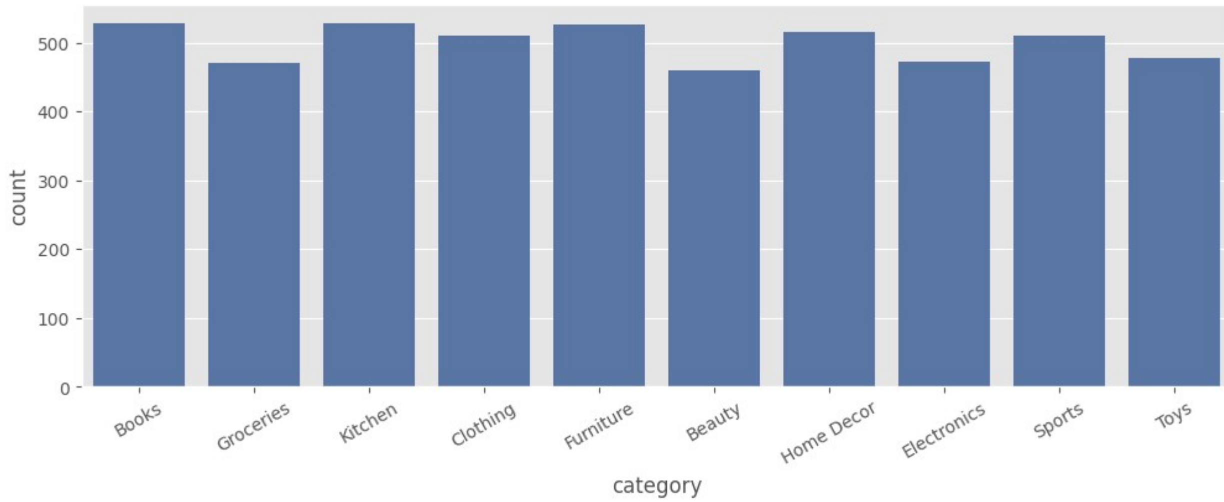



```
#Insight
#Most orders are low-value, Few orders are high-value & Right-skewed distribution

#Check Category Frequency
df['category'].value_counts()
```

```
category
Books          528
Kitchen        528
Furniture      527
Home Decor     515
Clothing       511
Sports         511
Toys           478
Electronics    472
Groceries      470
Beauty         460
Name: count, dtype: int64
```

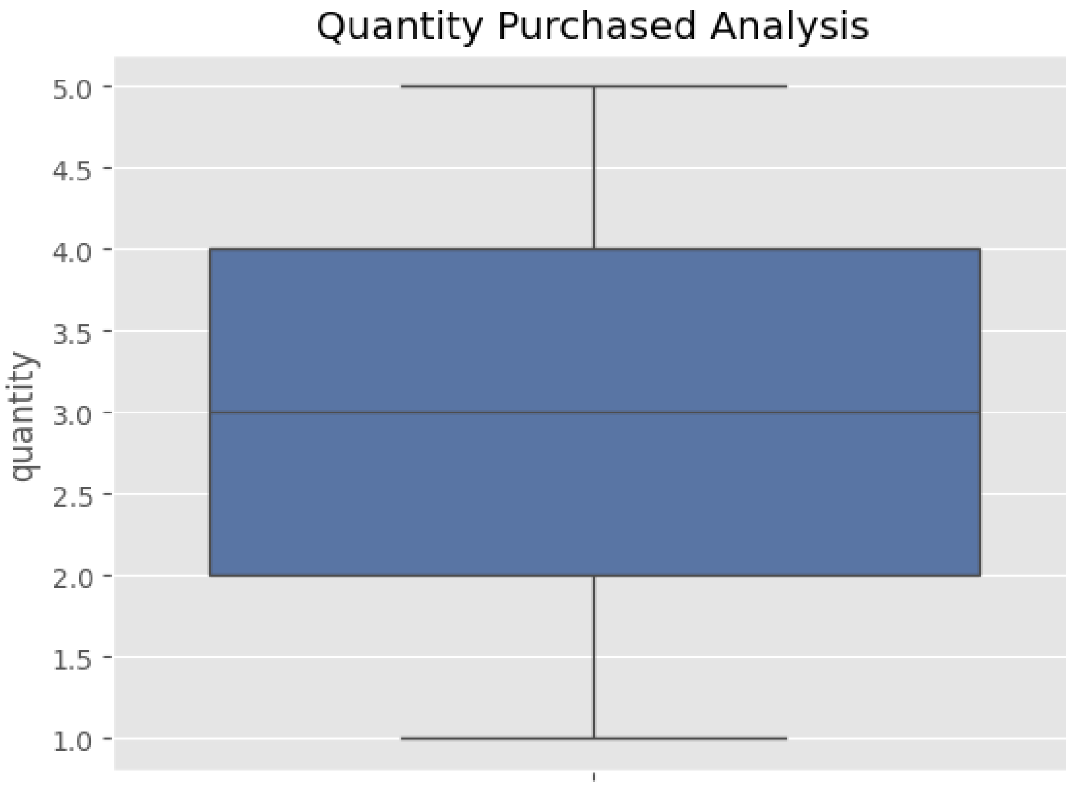
```
plt.figure(figsize=(12,4))
sns.countplot(x='category', data=df)
```

```
plt.xticks(rotation=30)
plt.show()
```

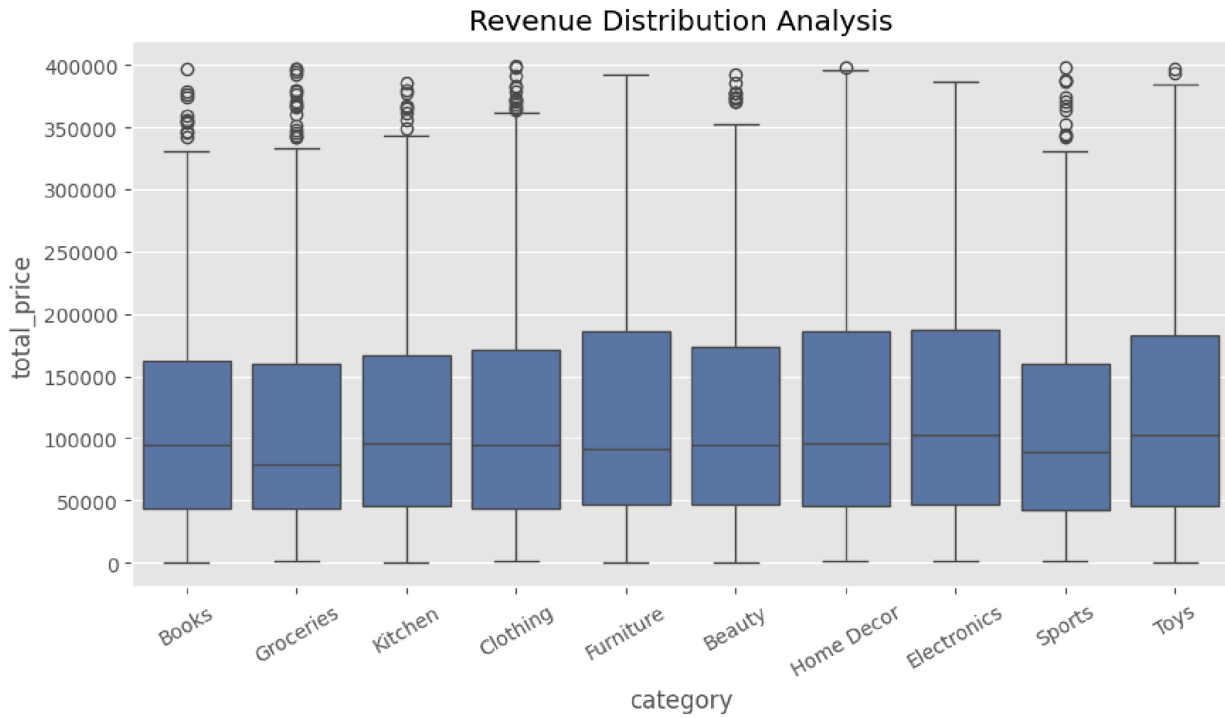



```
#Insight
#Shows which product categories dominate sales volume.

sns.boxplot(df['quantity'])
plt.title('Quantity Purchased Analysis')
```

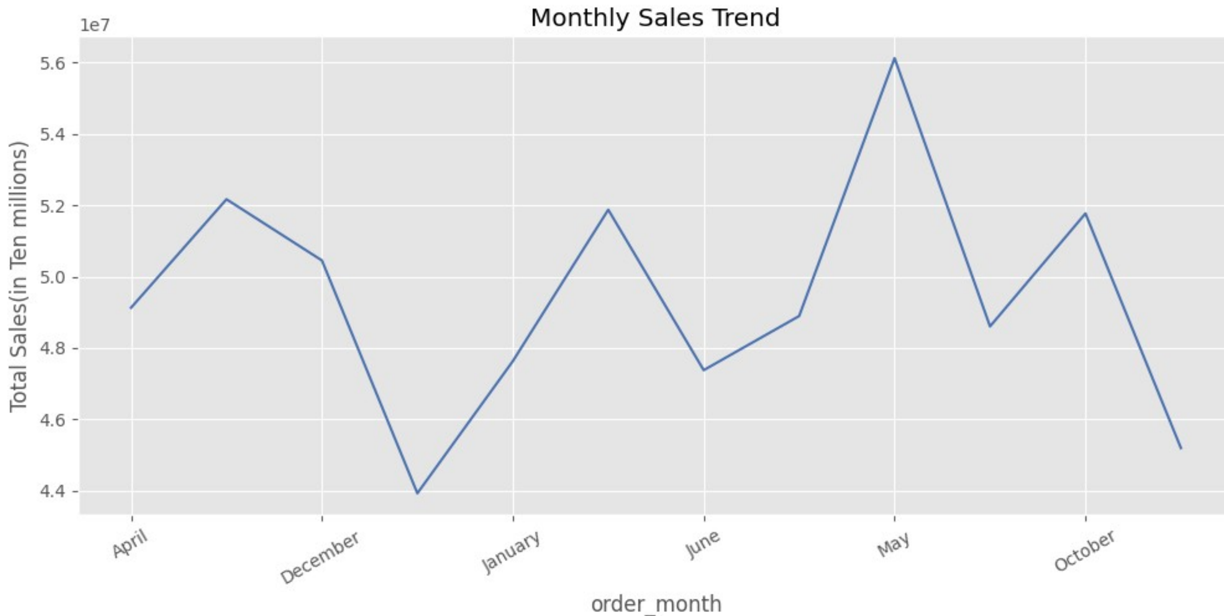
```
Text(0.5, 1.0, 'Quantity Purchased Analysis')
```



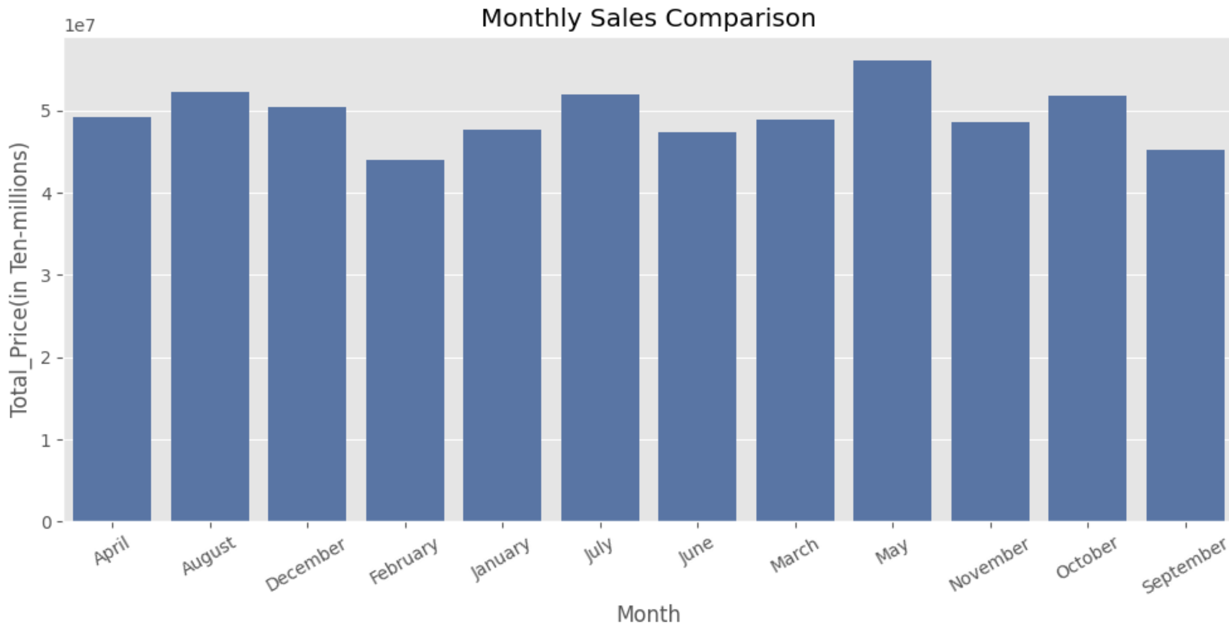

```
    #Insight
    #Most customers purchase products in small-to-medium quantities.
    #A small number of transactions contain significantly higher
quantities, creating visible outliers.

    # Multivariate Analysis

    plt.figure(figsize=(10,5))
    sns.boxplot(x='category', y='total_price', data=df)
    plt.title('Revenue Distribution Analysis')
    plt.xticks(rotation=30)
    plt.show()
```



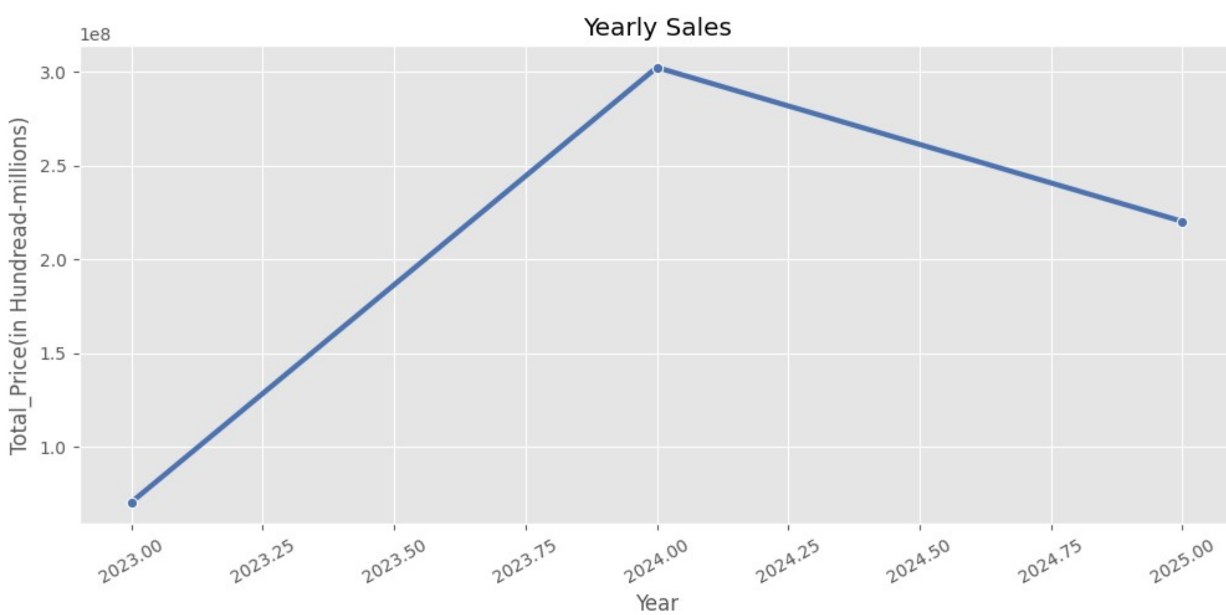

```
    #Insight
    #The majority of customers generate average transaction values
    #A smaller group of premium customers contributes a
disproportionately high share of revenue

    month_order = [
    'January','February','March','April','May','June',
    'July','August','September','October','November','December'
    ]
    monthly_sales = df.groupby('order_month')['total_price'].sum()
    monthly_sales.plot(figsize=(12,5))
    plt.title('Monthly Sales Trend')
    plt.ylabel('Total Sales(in Ten millions)')
    plt.xticks(rotation=30)
    plt.show()
```



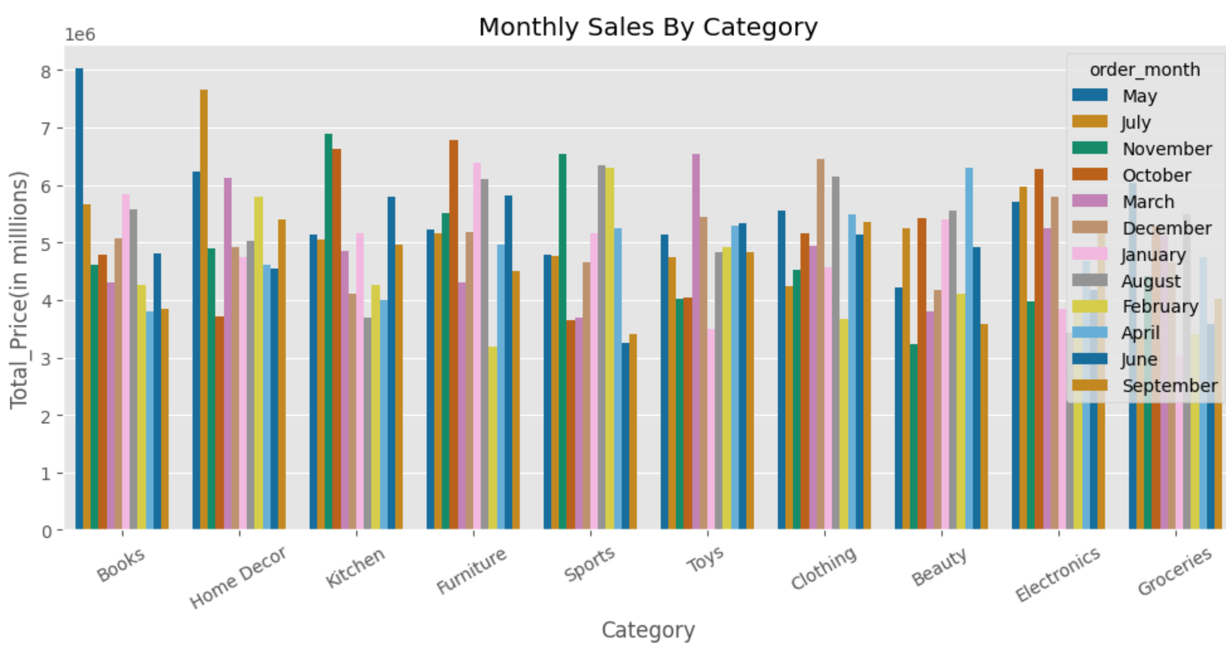

```
    #Insights
    #May recorded the highest monthly sales revenue
    #Certain months experienced lower revenue performance, indicating
seasonal demand variations

    monthly_sales = df.groupby('order_month')
['total_price'].sum().reset_index()
    plt.figure(figsize=(12,5))
    sns.barplot(x='order_month',y='total_price',data=monthly_sales)
    plt.title('Monthly Sales Comparison')
    plt.xlabel('Month')
    plt.ylabel('Total_Price(in Ten-millions)')
    plt.xticks(rotation=30)
    plt.show()
```




```
    yearly_sales = df.groupby('order_year')
['total_price'].sum().reset_index()
    plt.figure(figsize=(12,5))

sns.lineplot(x='order_year',y='total_price',data=yearly_sales,marker='
o', linewidth=3)
    plt.title('Yearly Sales')
    plt.xlabel('Year')
    plt.ylabel('Total_Price(in Hundread-millions)')
    plt.xticks(rotation=30)
    plt.show()
```

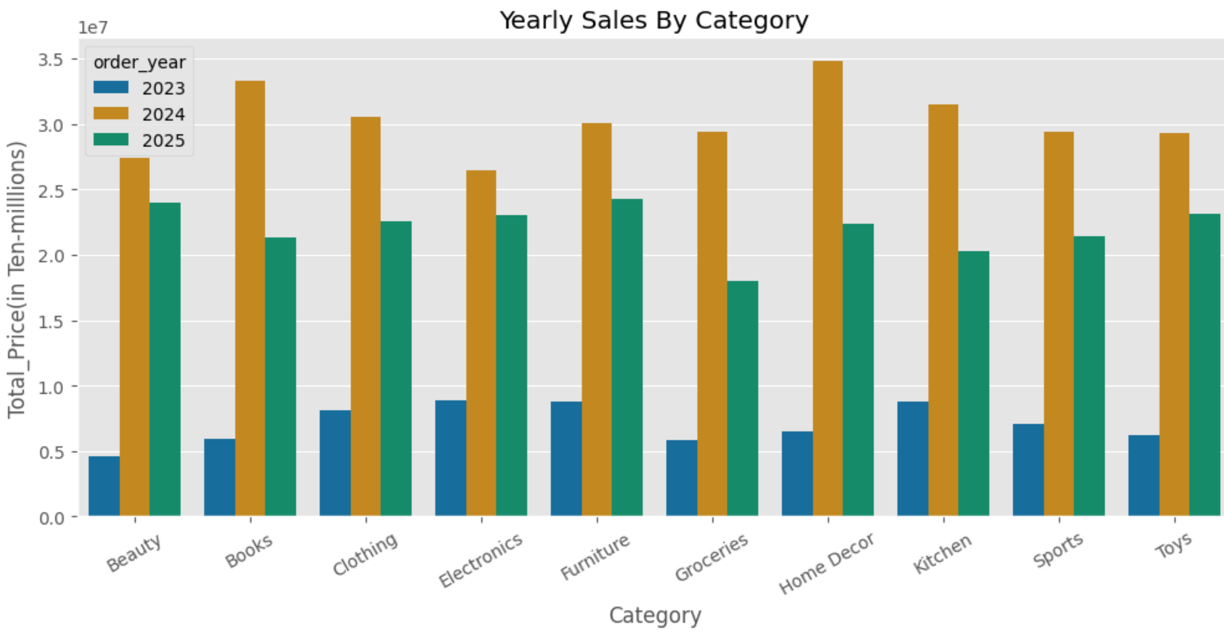


```
    #Insight
    #The yearly sales trend shows strong business growth from 2023 to 
2024, followed by a moderate decline in 2025,
    # while overall revenue remained significantly higher than the 
initial year.

    monthly_sales = df.groupby(['category','order_month'])
['total_price'].sum().reset_index()

monthly_sales=monthly_sales.sort_values('total_price',ascending=False)
    plt.figure(figsize=(12,5))

sns.barplot(x='category',y='total_price',data=monthly_sales,hue='order
_month',palette='colorblind',legend='brief')
    plt.title('Monthly Sales By Category')
    plt.xlabel('Category')
    plt.ylabel('Total_Price(in milllions)')
    plt.xticks(rotation=30)
    plt.show()
```

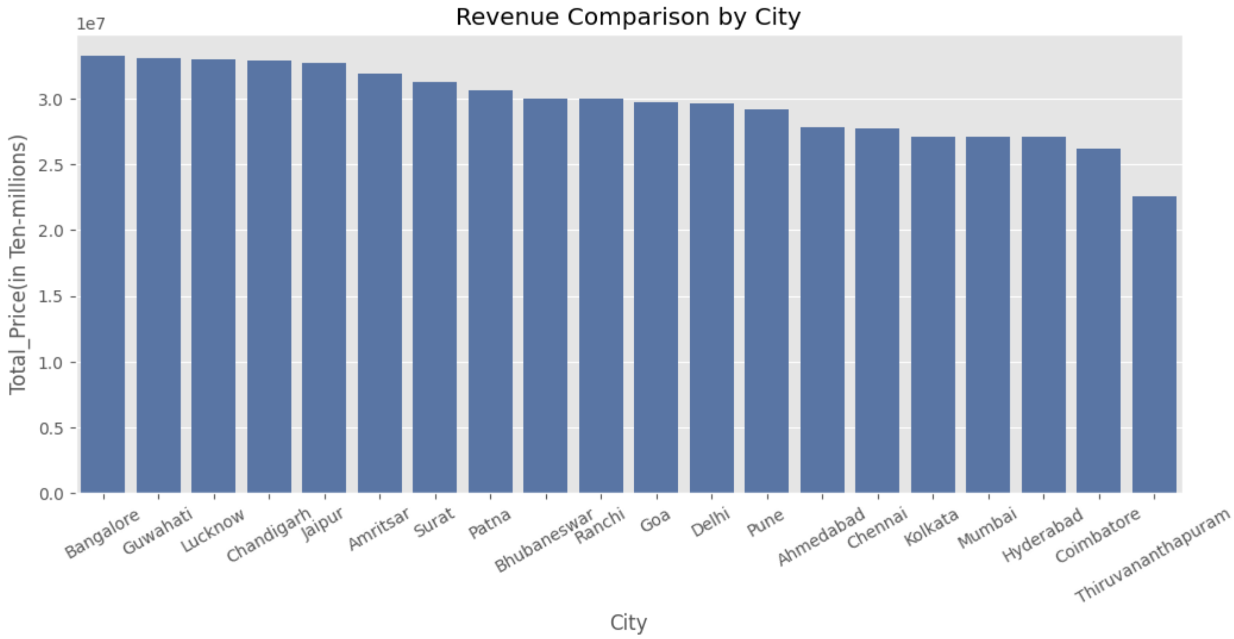



```
    #Insight
    # Book category contributes higher in revenue, closely followed by 
Home Decor category.

    plt.figure(figsize=(12,5))
    yearly_sales = df.groupby(['category','order_year'])
['total_price'].sum().reset_index()

sns.barplot(x='category',y='total_price',data=yearly_sales,hue='order_
year',palette='colorblind')
```

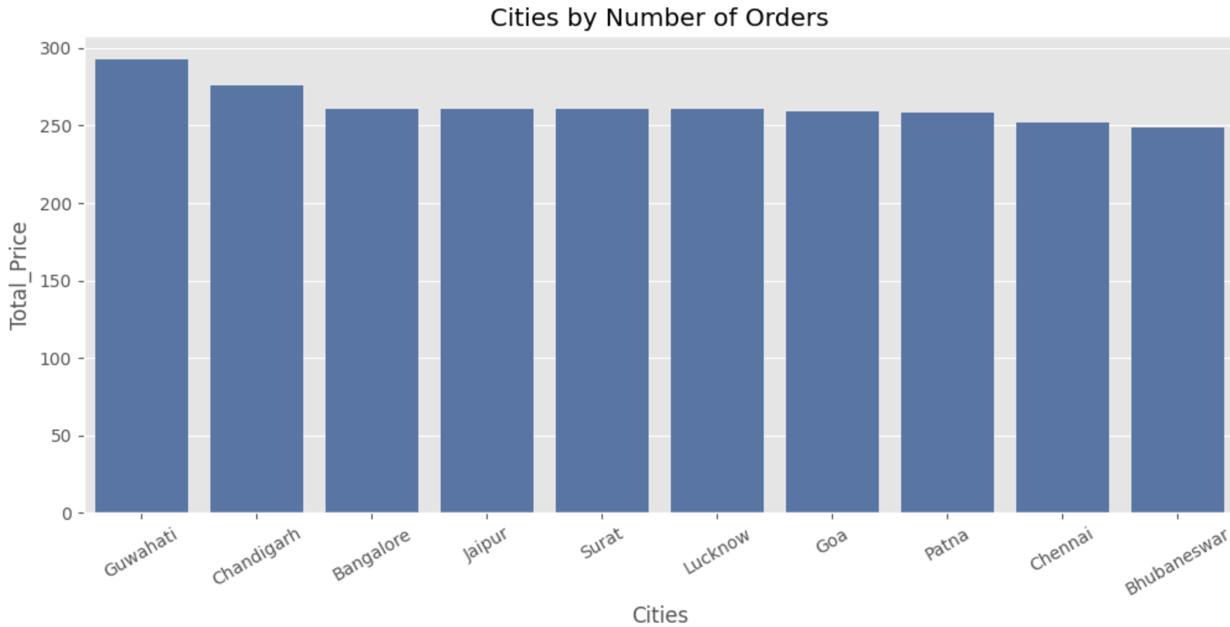
```
    plt.title('Yearly Sales By Category')
    plt.xlabel('Category')
    plt.ylabel('Total_Price(in Ten-milllions)')
    plt.xticks(rotation=30)
    plt.show()
```



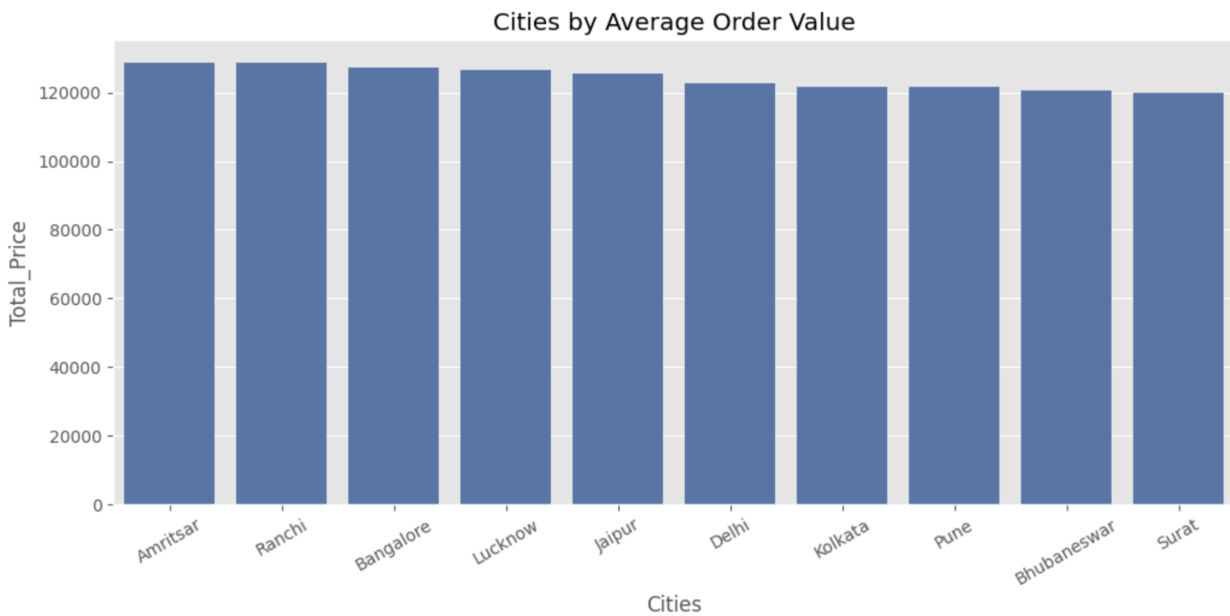

```
    #Insight
    #Sales peaked across almost all categories in 2024—especially Home
Decor and Books—while 2025 shows a noticeable decline,
    #with Groceries experiencing the sharpest drop.


    monthly_sales = df.groupby('city')
['total_price'].sum().reset_index()

monthly_sales=monthly_sales.sort_values('total_price',ascending=False)
    plt.figure(figsize=(12,5))
    sns.barplot(x='city',y='total_price',data=monthly_sales)
    plt.title('Revenue Comparison by City')
    plt.xlabel('City')
    plt.ylabel('Total_Price(in Ten-millions)')
    plt.xticks(rotation=30)
    plt.show()
```




```
    #Insight
    #Bangalore generated the highest sales revenue among all cities,
while Thiruvananthapuram recorded the lowest sales performance.
    #Overall, sales distribution across cities remained relatively
balanced, indicating stable market demand across multiple regions.

    plt.figure(figsize=(12,5))
    city_sales = df.groupby('city')
['total_price'].count().reset_index()
    city_sales = city_sales.sort_values('total_price',
ascending=False).head(10)
    sns.barplot(x='city',y='total_price',data=city_sales)
    plt.title('Cities by Number of Orders')
    plt.xlabel('Cities')
    plt.ylabel('Total_Price')
    plt.xticks(rotation=30)
    plt.show()
```

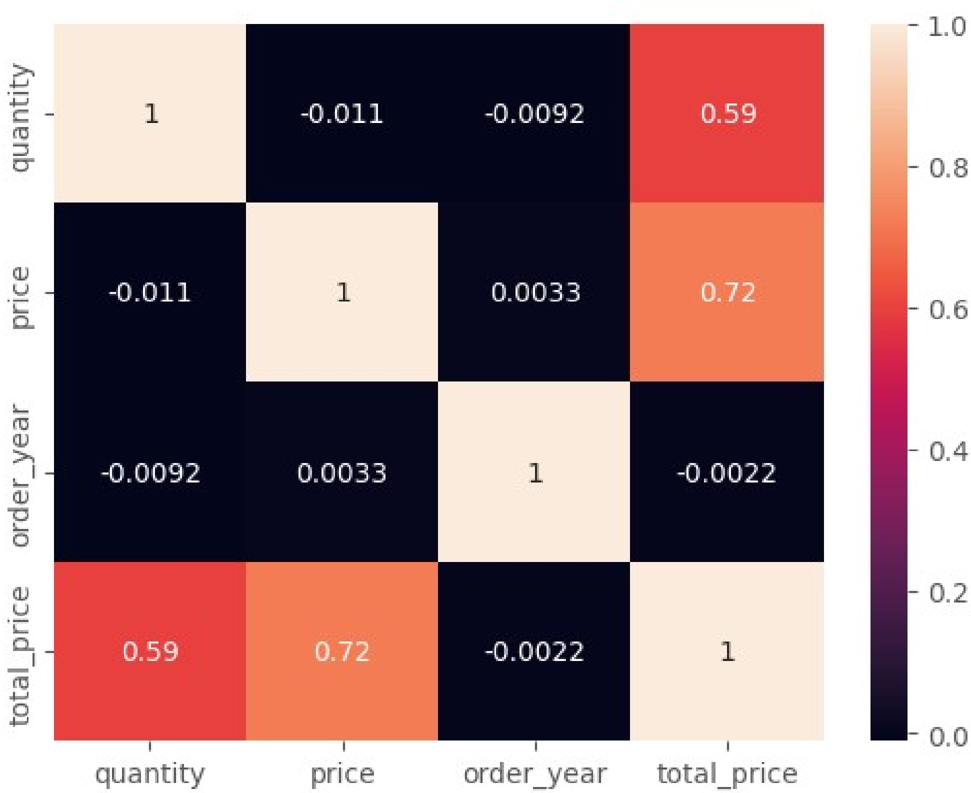



```
    #Insight
    #Guwahati shows higher number of orders,closely followed by
Chandigarh and other cities

    plt.figure(figsize=(12,5))
    city_sales = df.groupby('city')
['total_price'].mean().reset_index()
    city_sales = city_sales.sort_values('total_price',
ascending=False).head(10)
    sns.barplot(x='city',y='total_price',data=city_sales)
    plt.title('Cities by Average Order Value')
    plt.xlabel('Cities')
    plt.ylabel('Total_Price')
    plt.xticks(rotation=30)
    plt.show()
```




```
#Insight
#AOV of Top cities are mainly same

sns.heatmap(df.corr(numeric_only=True), annot=True)
```

```
<Axes: >
```

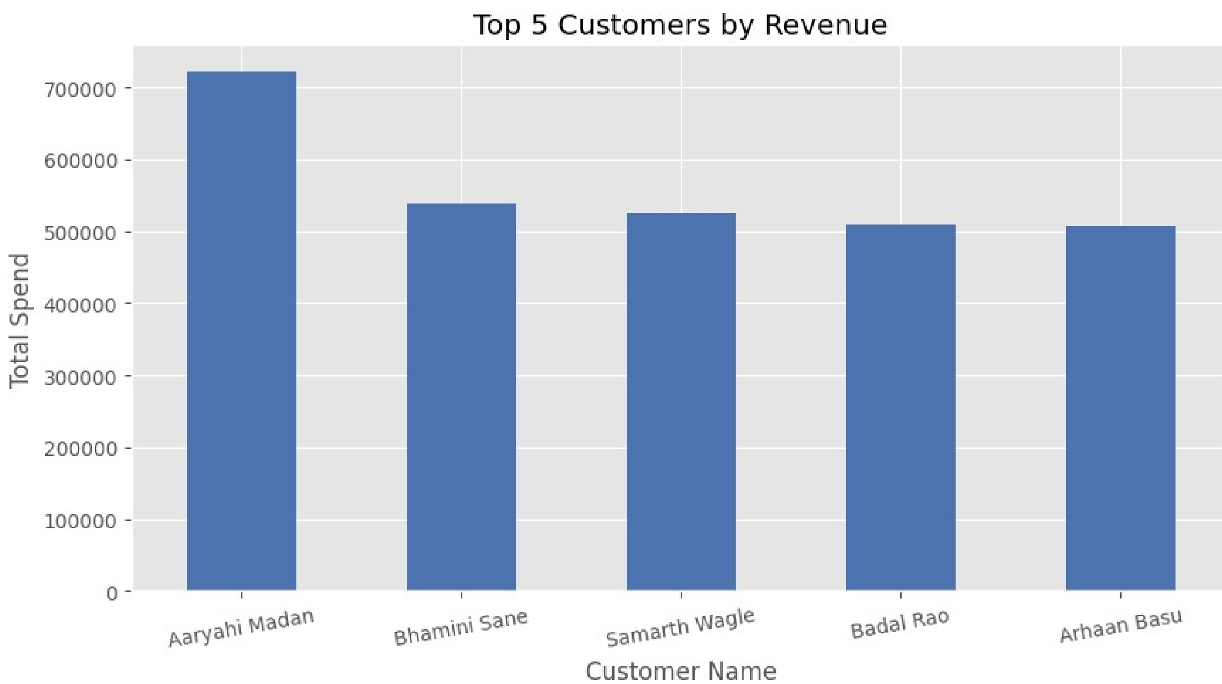


```
    #Insight
    #Total price is strongly positively correlated with price (0.72)
and moderately with quantity (0.59),
    #while order year shows almost no relationship with the other
variables.
```

Business KPIs

```
#Total Revenue
Total_Revenue=df['total_price'].sum()
print(Total_Revenue)

593171089

#Total Order
df.shape[0]

5000

#Average Order Value
AOV=df['total_price'].mean()
print(AOV)

118634.2178

# Top City in Sales

Top_City=df.groupby('city')['total_price'].sum().idxmax()
print(Top_City)

Bangalore

# Top Month in Sales
Top_Month=df.groupby('order_month')['total_price'].sum().idxmax()
print(Top_Month)

May

#Top 5 Customers in Spending
Top_Customers=df.groupby('customer_name')
['total_price'].sum().sort_values(ascending=False).head(5)
print(Top_Customers)

customer_name
Aaryahi Madan    722391
Bhamini Sane     538161
Samarth Wagle    524785
Badal Rao        509935
Arhaan Basu      506272
Name: total_price, dtype: int64
```

```python
top_customers = df.groupby('customer_name')
['total_price'].sum().sort_values(ascending=False).head(5)

top_customers.plot(kind='bar', figsize=(10,5))

plt.title('Top 5 Customers by Revenue')
plt.xlabel('Customer Name')
plt.ylabel('Total Spend')
plt.xticks(rotation=10)

plt.show()
```



Final Conclusion

Recommended Business Actions